[TODO: title] — Near Real-Time Markerless Scoring of the Standardized Drinking Task

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***Abstract*—Optical motion capture with a fitted biomechanical model is the reference standard for upper-limb movement quality after stroke, but it is lab-bound, and the markerless pipelines that now approach its accuracy need days of GPU optimisation per cohort, so a result never reaches the clinician who ordered it. We present a markerless pipeline that scores the standardized drinking task from multi-camera video — 2D pose per frame, a detect-once visual tracker for the cup, robust bundle adjustment, temporal smoothing, and movement-quality measures computed geometrically from 3D keypoints, with no inverse kinematics and no biomechanical model — and that segments the task into its phases from the cup and hand trajectories rather than from the mocap. On 750 trials from eleven recording units of a cohort recorded simultaneously with optical mocap, ten of the twelve drinking-task measures agree with the optical reference at $r_s \geq 0.84$ and $r_{av} \geq 0.93$ when the markerless recording supplies both the pose and the phase windows, and eleven of the twelve when both systems are given the same optical windows. A 9.2 s trial from five cameras is scored in 16.8 s on one consumer GPU, against roughly fifteen minutes per trial for the differentiable-biomechanics pipeline it is compared against. All but 1.3 s of that is per-frame work that a front end keeping pace with capture would absorb, which this card achieves at 30 Hz.**

***Index Terms*—markerless motion capture, optical motion capture, drinking task, movement quality, stroke rehabilitation, real-time**

I. Introduction

Upper-limb impairment persists in a large fraction of stroke survivors, and rehabilitation depends on knowing whether a movement is improving in quality and not merely in outcome. Clinical scales record whether a task was completed, not how, and are too coarse to track the compensations — extra trunk motion, shoulder abduction substituting for elbow extension — that distinguish recovery from adaptation. Kinematic analysis of the standardized drinking task [1] supplies that detail: movement time, smoothness, peak velocity and its timing, joint angles, trunk displacement and interjoint coordination summarise a single reach-to-drink into numbers with established clinical meaning. Obtaining them has meant optical motion capture — markers placed by a trained operator, a calibrated laboratory, and a biomechanical model fitted offline.

Markerless motion capture removes the markers and the operator, and has now closed much of the accuracy gap [2]. What it has not closed is time. The end-to-end differentiable-biomechanics pipeline of [2] required roughly twelve days on an RTX 3080 for 1160 trials, dominated by keypoint extraction over 5800 videos, with a further thirty CPU-hours of inverse kinematics on the optical side. The clinical value of these measures lies in feeding them back while the patient is still in the room.

We take the same task, measures and validation protocol, but choose every stage so that per-trial compute is bounded and overlaps with recording. Measures are computed geometrically from 3D keypoints rather than from a fitted biomechanical model, which costs some trajectory fidelity and buys a per-trial cost bounded by the recording itself.

Contributions:

- A markerless pipeline that produces the drinking-task movement-quality measures in near real time on a single consumer GPU, with a measured per-stage latency budget: a 9.2 s five-camera trial is scored in 16.8 s, of which only 1.3 s is work that cannot begin until the last frame.
- A validation against synchronized optical mocap on eleven recording units, reporting per-trajectory agreement on 790 trials and per-measure agreement on the 750 the segmenter accepts (Tables I–II).
- A phase segmentation driven by the markerless cup and hand rather than by the mocap, validated separately, whose failures are self-detected and therefore reportable in deployment — closing the standalone phase classification that [2] leaves open.
- An account of which measures survive the no-IK, geometric construction and which do not, and of how much of the residual disagreement is landmark correspondence rather than pose error.

II. Related Work

Movement quality in the drinking task. Alt Murphy et al. [1] identified the kinematic variables used here on the drinking task, showing that they separate stroke survivors from controls and grade mild from moderate impairment; the measurement protocol that defines them followed [3]. All of them presuppose a segmentation of the trial into phases, and the reference protocol [3] derives it from the mocap: two of its nine markers are placed on the glass, and the boundaries come from glass velocity crossing 15 and 10 mm/s, hand velocity crossing 2% of its own peak subject to a 20 mm/s floor, and a face-to-glass distance read against its own steady state, with recordings the detector mis-segments corrected by manual inspection.

Markerless capture for clinical kinematics. Markerless capture has been proposed for clinical measurement in rehabilitation for years, but a systematic review finds its actual application still preliminary and its benefits for assessing patients inconclusive [4]. Multi-view pose estimation followed by triangulation now recovers upper-limb motion accurately enough for clinical measures, and the remaining gap to optical capture is increasingly one of definition rather than of detection. Unger et al. [2] report the closest comparison to ours: the same task and measures, an end-to-end differentiable pipeline that fits a musculoskeletal model to the video in a physics engine [5], validated against simultaneous optical capture processed in OpenSim from a full-body model [6]. Their agreement is the standard we compare against, and their two weakest measures are the two weakest here. They classify the phases for both systems from the optical markers, which isolates pose error from segmentation but leaves markerless phase classification, in their words, still requiring validation for standalone application.

Latency. Processing time is rarely reported in this literature, and almost never per trial, which makes pipelines hard to compare and easy to mistake for deployable. We report it per stage and per trial, separating the part that can overlap recording from the part that cannot.

III. Methods

A. Dataset

We use the DELTA dataset of [2] unchanged — 40 repetitions of the standardized drinking task [1] per arm, recorded simultaneously by optical mocap at 100 Hz and by five synchronized webcams at 1080p and 60 Hz — and report here only the selection applied on top of it.

Participants. Ten participants from the same pool as [2]. One was recorded in two sessions under different calibrations, which we treat separately, giving eleven recording units. Recruitment, inclusion criteria and ethics approval are as reported in [2].

Cameras. Of the 55 unit–camera pairs, twelve were excluded: five whose clips were consistently mis-cut, and seven that were miscalibrated. Both faults were identified from the recordings themselves, independently of any comparison against the optical reference. Three of the eleven units retain all five cameras and the rest three or four.

Trials. Clips are paired with the resampled mocap over a common window, then admitted by two automatic gates: agreement between the two systems at the best lag, and the fraction of frames carrying a usable wrist. Where the recording stops before the hand has finished moving, total movement time is a lower bound and is withheld; every measure windowed on reaching keeps the full set.

B. Pipeline

The pipeline runs in six stages.

1) *2D body pose.* Stock `yolo26s-pose` [7], run per frame on every camera.
2) *Cup localization.* Stock `yolo26x-seg` [7] locates the cup *once*, at the first frame whose detections agree across at least three cameras; a transformer visual-object tracker (UETrack-B [8]) then follows it through the remaining frames in each camera. It is never re-anchored. The cup trajectory is used *only* to segment the movement into phases (Sec. III-C).
3) *3D reconstruction.* A multi-view DLT initialises a bundle adjustment [9] that minimises geometric reprojection error over all nine upper-body joints, weighted by detector confidence and robustified by a Huber kernel. A second term penalises the frame-to-frame *variance* of the five near-rigid segment lengths at a weight of 0.05 relative to the reprojection term. The results below are unchanged for weights between 0.025 and 0.2, while removing the term costs 0.03 on peak velocity and 0.02 on peak elbow angular velocity. No camera is excluded from the body fit; a bad view is down-weighted rather than rejected. Gating happens only at initialisation, where cameras reprojecting beyond 30 px are dropped from the DLT. Below two cameras the frame is left empty, which happens on 0.6% of joint-frames; those frames stay empty downstream and are excluded from the reductions that read them rather than interpolated. The cup is reconstructed by the same solve but under a stricter gate: frames with fewer than three agreeing cameras are discarded rather than forced, and interior gaps shorter than 0.75 s are filled by a Kalman smoother.
4) *Temporal smoothing.* SmoothNet [10], applied to the 3D joint trajectories.
5) *Phase segmentation.* From the cup and hand trajectories; see Sec. III-C.
6) *Measure extraction.* Geometric, from 3D body keypoints only; see Sec. III-D.

C. Phase Segmentation

Segmentation reduces to a single primitive applied to two distance channels: a distance closes, holds flat, then opens again. On the wrist-to-cup distance that pattern is grasp and release; on the hand-to-mouth distance it is the drinking phase. Reach onset is referenced to the hand's resting position at the start of the trial, the final settle to where it comes to rest at the end.

A closing or opening run counts only if it traverses a fixed fraction of that channel's own range within the trial; boundary agreement is unchanged for fractions anywhere between 0.2 and 0.45. Boundaries are found in one forward pass, each taken as the first qualifying run after the previous boundary. What distinguishes this from the reference protocol [3] is that the cup is followed visually rather than instrumented with markers; both rules otherwise mix absolute thresholds with thresholds relative to the trial's own range. Where no usable cup track exists the segmenter declines rather than returning a boundary, which is reported in Sec. IV-E.

D. Movement Quality Measures

Measures are computed geometrically from the 3D keypoints; no biomechanical model is fitted and no inverse kinematics is solved. Throughout, position is smoothed and then differentiated, never the reverse. Movement units are counted against a 60 mm/s amplitude rather than the 20 mm/s of marker-based practice, the markerless wrist moving at 13 mm/s at rest against 4 mm/s for the optical markers; agreement is insensitive to the choice, r_s moving between 0.80 and 0.82 as the threshold is swept from 20 to 80 mm/s on both systems. The remaining constants are in the released code.[1]

The twelve measures are those of the drinking-task protocol [3]; only the shoulder angles need defining here. An orthonormal body frame is built from four torso points — both shoulders and both hips — giving a down axis from the shoulder midpoint to the hip midpoint, a frontal-plane normal, and a lateral axis, and is frozen per trial. Shoulder flexion is the angle of the upper arm from the down axis within the sagittal plane, abduction the angle within the frontal plane, following that protocol's planar definitions.

Trunk displacement in [2] follows a sternum marker, which the COCO keypoint set does not contain, so ours is the three-dimensional excursion of the shoulder midpoint from its resting position rather than an anatomical forward projection.

Peak velocity is the one exception to the smoothing rule: it differentiates the pose without the additional position low-pass and low-passes the resulting speed instead.

Shoulder flexion is reported over two windows, as in [2]: from reach onset to the end of the drinking phase, and over the drinking phase alone. The maximum is taken within each.

E. Latency Model

We report processing time per trial, per stage and in total, on stated hardware. Two intervals are distinguished: the total, and the part that cannot begin until the recording ends. The stages that consume raw frames — pose estimation and cup tracking — need only the frame in hand and are separated from those that need the whole trial, so the second interval is the latency a deployment with a live front end would show.

Recording and inference do not share the GPU here: the clips are recorded first and processed afterwards. Video decoding is measured but excluded from the reported times, the archived clips being an artefact of this dataset rather than of the method.

[1] https://github.com/foalengtoussaint/fastmmc

F. Validation Protocol

All measures are computed from our markerless keypoints alone; the optical system contributes only the values compared against, computed from the optical markers using the same definitions (Sec. III-D), so the comparison isolates pose error rather than a difference of definition.

A marker on the skin and a detected keypoint label different points on the body, so a constant offset between the two is fitted per landmark, in the body and arm frames, and applied before the measures are compared. The shoulder, elbow and wrist offsets are held in a body frame frozen per trial; the sternum's in one rebuilt per frame. This is a landmark-matching step and not a correction of our estimate: neither system is positionally authoritative. The offset is fitted per participant and arm, and validated on trials held out from the fit. Its magnitude, and what happens when it is pooled across participants instead, are reported in Sec. IV-C.

The two systems are aligned in time by the same multi-signal criterion that admits a trial. For the per-frame trajectory comparison a static offset is estimated per trial and reported (Table I), and removed before RMSE, as in [2]; Pearson r is computed on the raw series. Per-measure agreement is Pearson r_s over single trials and r_{av} over per-participant, per-arm averages, following [2].

IV. Results

A. Study Population

[TODO: Study population. THREE blockers, established 2026-08-25. (a) FMA-UE per participant is NOT recoverable from this machine or the institutional share: the DELTA notebook computes it from a CSV on someone's local disk and emits only a bar chart, so the totals exist as pixels, not numbers, and the share carries no clinical export (searched 2026-08-25). This is the measure Sec. VI's "impairment gradient is mild" rests on. (b) Age, sex, dominant hand, affected side, stroke date and Box-and-Block ARE recoverable, but keyed by DELTA Record ID with no key to the P-labels on this machine; assuming Record ID equals the P-number matches the affected side on 8 of 9 informative units and CONTRADICTS on P25, so it cannot be used. (c) An explicit ethics and consent statement is still needed rather than the deferral to [2] in Sec. III-A — confirm whether BASEC-No 2022-00491 covers this re-analysis. Needed to close (a) and (b): the FMA-UE totals in any tabular form, and the Record ID to P-label key or P25's affected side. Trial counts per unit and arm are already in VERIFY.md.]

The two admission gates of Sec. III-A pass 842 paired clips.[2] Of these, 45 lack a usable optical wrist or nose marker, three are mispaired between the systems, and four are withdrawn where an optical marker jumps and does not return, leaving 790: the trial set of Table I. One trial yields

[2] The scripts that produce every table and figure below, and the record of what each number was measured from, are at https://github.com/foalengtoussaint/cup-task.

no segmentation and the segmenter declines a further 39 (Sec. IV-E), leaving 750 for Table II.

On 146 of the 750 trials (19.5%) the recording stops before the hand comes to rest, and total movement time is withheld on those; with two further trials carrying no usable value, it is reported on $n = 602$.

B. Kinematic Trajectory Agreement

Table I reports per-frame agreement for the six kinematic trajectories over all 790 trials of the eleven recording units (413 affected, 377 unaffected), and Fig. 1 shows exemplary trajectories for one participant. The shape of the optical trajectory is recovered closely: elbow extension at $r = 1.00$ on both arms with an RMSE of 4.2 deg, shoulder flexion at 0.96 and 0.97, the velocity trajectories at 0.96–0.98, and trunk displacement at 0.96 and 0.98. Shoulder abduction is the weakest (0.78 unaffected, 0.88 affected) and the only trajectory worse on the unaffected arm. The median lag between the systems is one video frame (0.017 s) on every trajectory.

Per-frame correlation is a generous statistic here, because both systems watch the same stereotyped movement: pairing a markerless trajectory with the optical trajectory of a *different* trial by the same participant and arm still gives $r = 0.93$ on elbow extension and 0.85 on shoulder flexion. The velocity trajectories are far less forgiving under that null (0.67 and 0.65), as is abduction (0.53), so it is the RMSE and bias columns that carry what the correlation does not.

The offsets removed before the RMSE column are not small. Taking the offset as markerless minus optical, they are +7.4 and +8.1 deg on elbow extension, −5.0 and −3.2 deg on shoulder flexion, and −6.0 and −2.0 deg on shoulder abduction: the markerless elbow reads more extended than the optical one and the markerless shoulder less flexed and less abducted, on both arms. The last row of each block in Table I reports how much each varies within a participant, and the answer is roughly a degree: 1.1 deg for elbow extension and shoulder flexion, 0.8–0.9 deg for abduction, 2.8 mm for trunk displacement. A discrepancy of several degrees whose trial-to-trial spread is one degree is a constant of the recording session rather than a property of the estimate, which is what the next section quantifies.

C. Landmark Correspondence

The offset carries into the measures. Taking the per-trial difference between the markerless and the optical value of each maximum angle and grouping the 750 trials by participant and arm (21 groups), the group means span −19.4 to +9.8 deg for shoulder flexion, +1.3 to +19.2 deg for elbow extension and −15.0 to +3.1 deg for shoulder abduction, while the median within-group standard deviation is 1.3 to 1.7 deg. Between-group differences account for 79% to 93% of the variance of the per-trial difference: each recording session contributes a nearly fixed angular offset, and it differs by session. Correlating the uncorrected measures within each participant and arm rather than across the cohort makes the same point — shoulder flexion reaches 0.86 that way against

TABLE I
MEDIANS AND [IQRS] ACROSS TRIALS FOR THE SIX KINEMATIC TRAJECTORIES, BY ARM. BIAS IS MARKERLESS MINUS OPTICAL AND IS REMOVED BEFORE RMSE; THE BIAS IQR ROW IS THE MEAN ACROSS PARTICIPANTS OF EACH PARTICIPANT'S OWN IQR OF IT.

Kinematic	Value	Unaffected arm	Affected arm
End-effector vel.	r	0.98 [0.98, 0.99]	0.98 [0.97, 0.98]
	RMSE (mm/s)	37.01 [28.76, 45.03]	42.51 [33.68, 48.42]
	Bias (mm/s)	-5.68 [-8.62, -1.69]	-8.91 [-12.59, -3.65]
	Bias IQR (mm/s)	4.54	4.76
Elbow ang. vel.	r	0.97 [0.96, 0.98]	0.96 [0.95, 0.97]
	RMSE (deg/s)	11.70 [9.79, 13.90]	11.84 [10.07, 13.63]
	Bias (deg/s)	4.03 [3.01, 5.06]	3.84 [2.67, 5.59]
	Bias IQR (deg/s)	1.17	1.11
Elbow extension	r	1.00 [0.99, 1.00]	1.00 [0.99, 1.00]
	RMSE (deg)	4.16 [3.65, 4.73]	4.24 [3.24, 5.15]
	Bias (deg)	7.44 [4.61, 8.98]	8.08 [3.05, 11.07]
	Bias IQR (deg)	1.08	1.06
Shoulder flexion	r	0.96 [0.92, 0.98]	0.97 [0.93, 0.98]
	RMSE (deg)	3.26 [2.59, 4.07]	2.96 [1.65, 3.97]
	Bias (deg)	-5.04 [-8.78, -3.47]	-3.17 [-5.99, -0.91]
	Bias IQR (deg)	1.17	1.07
Shoulder abduction	r	0.78 [0.62, 0.91]	0.88 [0.77, 0.94]
	RMSE (deg)	2.33 [1.79, 4.16]	2.19 [1.74, 3.06]
	Bias (deg)	-5.98 [-7.95, -2.88]	-1.95 [-7.33, 0.09]
	Bias IQR (deg)	0.84	0.94
Trunk displacement	r	0.96 [0.89, 0.98]	0.98 [0.94, 0.99]
	RMSE (mm)	3.96 [3.14, 6.11]	4.50 [3.73, 6.77]
	Bias (mm)	1.68 [-0.50, 3.67]	0.71 [-2.70, 4.06]
	Bias IQR (mm)	2.81	2.77

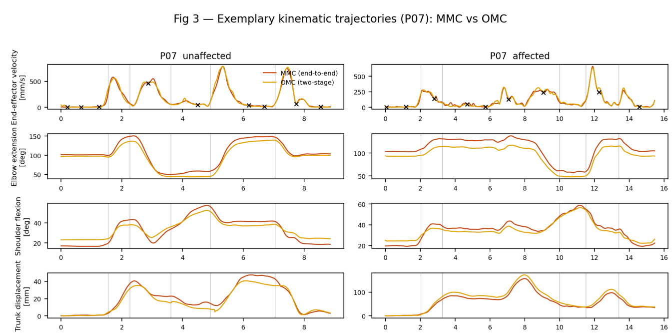

Fig. 1. Exemplary kinematic trajectories for the affected and unaffected arm, MMC versus OMC, with drinking-task phases marked and movement units indicated on the end-effector velocity. Participant P07, trials 10 (left, unaffected) and 43 (right, affected).

the 0.68 obtained by pooling — and the fitted offsets leave that within-session figure unchanged at 0.85.

Fitting the offsets of Sec. III-F against that difference recovers displacements largest where marker and keypoint are anatomically furthest apart: 48 mm at the shoulder in the trunk frame and 97 mm in the segment frame, 41 mm at the elbow, and 2.9 ± 6.8 mm at the wrist. Each offset's standard deviation across the 21 groups is comparable to its mean, making it a per-session quantity. Within a participant and arm the match generalises to held-out trials, agreement falling within 0.007 of the in-sample value; fitted on every other participant and applied to the held-out one it does not, shoulder flexion falling below its uncorrected value (0.72 to 0.55) while the others move by less than 0.02. The disagreement is thus not a fixed difference between the optical marker set and the detected keypoints, which would transfer, but specific to the session.

The match acts on the angles and on what is derived from them. Against the same table computed without it, shoulder flexion rises by 0.28 (r_s 0.68 to 0.97), its drinking-phase counterpart by 0.24, elbow extension and abduction by 0.08 each, interjoint coordination — a correlation between two of those angles — by 0.08, and peak elbow angular velocity, differentiated from the elbow angle, by 0.02, while peak velocity, both time-to-peak measures, movement units and total movement time move by 0.003 or less. The trunk takes the same treatment: the optical sternum and the markerless shoulder midpoint sweep different arcs during a forward lean, and offsets of 54 to 296 mm lift the measure from $r_s = 0.93$ to 0.95 and $r_{av} = 0.96$ to 0.99.

D. Movement Quality Measures

Table II reports agreement for the twelve movement-quality measures with both systems given the same optical phase windows, which isolates pose error from segmentation. Eleven of the twelve reach $r_s \geq 0.84$ and $r_{av} \geq 0.93$. The four maximum joint angles sit at 0.94–0.98 after landmark matching, peak velocity and peak elbow angular velocity at 0.89 and 0.91, trunk displacement at 0.95, and the two time-to-peak measures at 0.99 and 1.00. Number of movement units, a count of speed oscillations exceeding an amplitude threshold, is the lowest of the eleven at 0.84, rising to 0.93 on per-participant averages. The two shoulder-flexion measures agree equally well over their different windows, 0.97 over reach onset to the end of drinking and 0.97 over the drinking phase alone, the latter on the 748 trials where a drinking window exists.

Total movement time is derived entirely from the phase boundaries, which both systems share here, so its $r_s = 1.00$ is a consistency check rather than evidence about pose; it is informative only in the markerless-window column.

Interjoint coordination is the exception at $r_s = 0.48$, against $r_{av} = 0.83$, and the gap is one of range rather than of error: in a cohort with largely intact coordination its reference values are compressed near the top of their range, 18 of the 747 trials (2.4%) carrying 92% of the variance of the optical value (Sec. V).

E. Phase Segmentation

Segmenting from the cup is validated separately: first the boundaries themselves, then what the measures cost when the windows come from the markerless recording.

The segmenter declines 39 of 790 trials (4.9%). On these no cup frame survives the three-camera floor, the wrist-to-cup distance is flat, and the grasp collapses onto its floor at reach onset. The failures concentrate where the cup has no redundancy: 12.9% of trials at the four units with three cameras against 0.4% at the rest, the floor there requiring every camera to agree.

Table III reports boundary agreement on the remaining 750 trials, as the disagreement between the same rule run on the markerless and on the optical cup, paired per trial. Four of the six boundaries agree to within 50 ms at the median, reach onset and settle to within 17 ms. Grasp is the weakest at 100 ms, followed by drink onset at 67 ms.

That comparison quantifies sensitivity to the input, not agreement with a different rule. For the latter the dataset carries the boundaries used by [2], its own implementation of the reference protocol [3]: hand velocity above 20 mm/s held for 0.3 s for reach onset, glass velocity above 50 mm/s for grasp, a hand-to-face distance crossing 1.2 times its own steady-state value for the drink boundaries, and hand velocity falling below 2% of its peak for the settle — the reference protocol's own mix of absolute and relative rules at different constants, still instrumented on the glass. Those boundaries exist for 631 of the 750 trials; the remaining 119 carry no reference row, that detector having produced no phase set for them. Against the 631, our rule run on the optical cup differs by a constant per boundary — +200 ms on reach onset, −150 on grasp, +167 on drink onset, −100 on drink offset — with a residual of 33 to 67 ms once that constant is removed. Release and settle differ more, by +483 and −433 ms with residuals of 233 and 117 ms. The two rules mark the same events by different criteria, theirs velocity crossings and ours displacement runs. The settle differs most and differs in kind: theirs waits for hand velocity to fall to 2% of its peak, while ours fires when the hand first reaches the neighbourhood of its final position, at that moment still moving at half its peak speed. Our windows are therefore systematically shorter — reaching by 31% (0.80 against 1.17 s), the whole movement by 10% — so absolute values of the window-dependent measures are not on the reference protocol's scale. The correlations in Table II are unaffected, both inputs being segmented by the same rule.

The markerless-window columns of Table II and Fig. 2 give the end-to-end result, with the markerless recording supplying both the pose and the windows. Five of the twelve show a degradation distinguishable from zero against the optical-window column, and all five are small: number of movement units by 0.036 (interval 0.019 to 0.066), time to peak velocity by 0.033, peak elbow angular velocity by 0.013, and peak velocity and total movement time by 0.005 each. The remaining seven cannot be distinguished from no change at all. Interjoint coordination appears to move most of all, improving by 0.088 to 0.57, but a paired bootstrap puts that change at $[-0.09, +0.21]$: its correlation is too unstable for a shift of that size to carry information. Elbow extension is unchanged by construction, being reduced over the whole movement rather than over a phase; trunk displacement is reduced over the same span and is likewise unchanged here, both boundaries of it being shared.

The pattern follows the windows rather than the boundary errors. Peak velocity, peak elbow angular velocity, both time-to-peak measures, movement units and interjoint coordination are all reduced over the reaching phase, which ends at the grasp — the weakest boundary — and those are the measures that move. The two drink boundaries, the next weakest, bound

TABLE II
MOVEMENT-QUALITY MEASURES, MMC VERSUS OMC: r_s OVER SINGLE TRIALS AND r_{av} OVER PER-PARTICIPANT, PER-ARM AVERAGES, WITH THE PHASE WINDOWS TAKEN FROM THE OPTICAL AND FROM THE MARKERLESS RECORDING IN TURN, AND THE MARKERLESS CONDITION REPEATED AT 30 HZ. BRACKETS ARE 95% BOOTSTRAP INTERVALS, RESAMPLING TRIALS FOR r_s AND THE 21 PARTICIPANT–ARM GROUPS FOR r_{av}. n IS THE 60 HZ PAIR COUNT.

Measure	Optical windows		Markerless windows		30 Hz	
	r_s	r_{av}	r_s	r_{av}	r_s	n
PV [mm/s]	0.89 [0.88, 0.91]	0.94 [0.82, 0.98]	0.89 [0.87, 0.91]	0.94 [0.83, 0.98]	0.86 [0.83, 0.88]	750
Elbow angular PV [deg/s]	0.91 [0.89, 0.93]	0.97 [0.94, 0.99]	0.90 [0.88, 0.92]	0.98 [0.95, 0.99]	0.89 [0.85, 0.92]	750
Time to PV [s]	0.99 [0.99, 0.99]	0.99 [0.99, 1.00]	0.96 [0.90, 0.99]	0.99 [0.99, 1.00]	0.96 [0.90, 0.99]	748
Time to first PV [s]	1.00 [0.99, 1.00]	1.00 [0.99, 1.00]	1.00 [0.99, 1.00]	1.00 [0.99, 1.00]	0.99 [0.99, 1.00]	748
Movement units [n]	0.84 [0.75, 0.89]	0.93 [0.31, 0.98]	0.80 [0.69, 0.87]	0.89 [0.28, 0.96]	0.82 [0.72, 0.88]	748
Total movement time [s]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	0.99 [0.99, 1.00]	1.00 [0.99, 1.00]	0.99 [0.98, 1.00]	602
Interjoint coordination	0.48 [0.25, 0.82]	0.83 [0.34, 0.95]	0.57 [0.38, 0.82]	0.88 [0.25, 0.97]	0.30 [0.14, 0.72]	747
Trunk displacement [mm]	0.95 [0.93, 0.97]	0.99 [0.96, 1.00]	0.95 [0.93, 0.97]	0.99 [0.96, 1.00]	0.94 [0.91, 0.97]	750
Shoulder flexion [deg]	0.97 [0.96, 0.98]	0.99 [0.98, 1.00]	0.97 [0.96, 0.97]	0.99 [0.98, 1.00]	0.97 [0.96, 0.97]	750
Shoulder flexion, drink [deg]	0.97 [0.96, 0.98]	0.99 [0.98, 1.00]	0.97 [0.96, 0.98]	0.99 [0.98, 1.00]	0.97 [0.96, 0.98]	748
Elbow extension [deg]	0.98 [0.98, 0.98]	0.99 [0.98, 1.00]	0.98 [0.98, 0.98]	0.99 [0.98, 1.00]	0.98 [0.98, 0.98]	750
Shoulder abduction [deg]	0.94 [0.93, 0.96]	0.97 [0.92, 0.99]	0.94 [0.93, 0.96]	0.97 [0.91, 0.99]	0.94 [0.93, 0.96]	750

TABLE III
PHASE-BOUNDARY AGREEMENT BETWEEN MARKERLESS AND OPTICAL INPUT TO THE SAME SEGMENTER, PAIRED PER TRIAL.

Boundary	n	Median (ms)	p_{90} (ms)	> 0.25 s (%)
Reach onset	750	17	67	0.8
Grasp	750	100	233	8.3
Drink onset	748	67	233	9.1
Drink offset	748	50	200	4.1
Release	748	33	117	3.2
Settle	750	17	83	2.0

the shoulder angles, whose reduction is a maximum over a window and is insensitive to a 50–67 ms shift of either end: flexion and abduction move by 0.0002 or less, and drinking-phase flexion, reduced over the shortest window, by 0.003. The cost of self-segmentation is thus concentrated in the derivative measures of the reaching phase.

F. Latency and Capture Rate

On one consumer GPU (RTX 3060 Ti, 8 GB; i7-13700K), a trial of five cameras and 549 frames — 9.2 s of recording — is scored in 16.8 s; the excluded video decode costs a further 3.7 s. Pose estimation and cup tracking dominate: each must see every frame of every camera, and together they take 15.4 s, 28 ms per five-camera frame at 94% GPU utilisation. With the cup-seed scan that precedes them, 15.6 s of the total consumes single frames and everything after costs 1.3 s. That 15.6 s is 1.7 times the duration of the recording, so at five cameras and 60 Hz it cannot overlap capture on this card and the full 16.8 s follows the last frame. Where the front end does keep pace — three cameras, or 30 Hz capture, below — the single-frame stages run during the movement and the wait after it is the remaining 1.3 s. For scale, [2] reports about 15 minutes per trial on an RTX 3080, a card faster than the one used here, so the ratio is 53 per trial.

That cost scales with the number of cameras, and five is the worst case here: only three of the eleven units keep five, and the rest run three or four. Measured on one trial's frames held in memory, the two networks alone take 14.9 ms per rig-frame at three cameras, 19.1 at four and 24.7 at five — the 28 ms above being the same pair inside the running pipeline, which adds queueing and result parsing. So at three cameras the front end is at the 16.7 ms budget of 60 Hz capture and at four or five it is inside the 33.3 ms of 30 Hz. That is a statement about time alone: the body fit tolerates three cameras, but the cup's three-camera floor has no redundancy there (Sec. IV-E), so the rate and the camera count cannot both be reduced.

The 30 Hz column of Table II repeats the end-to-end comparison at half the capture rate on all 750 trials, with the temporal smoother's window scaled to the rate. The same ten of twelve measures clear $r_s \geq 0.84$ and $r_{av} \geq 0.93$ as at 60 Hz, and every measure stays within 1.5% of its 60 Hz value. Only peak velocity changes detectably, losing 0.028 (interval 0.011 to 0.051); the other eleven, interjoint coordination among them at 0.30 against 0.57, cannot be distinguished from no change at all.

V. DISCUSSION

Ten of the twelve movement-quality measures reach $r_s \geq 0.84$ and $r_{av} \geq 0.93$ end to end, without markers, without inverse kinematics and without a biomechanical model. The differentiable-biomechanics pipeline of [2] takes its phase windows from the optical markers for both systems, so the column of Table II comparable to it is the optical-window one. Two of their figures fall outside our bootstrap intervals: time to first peak velocity, 1.00 $[0.99, 1.00]$ against their 0.94, and peak elbow angular velocity, 0.91 $[0.89, 0.93]$ against 0.87. We are within 0.02 on the angles, trunk displacement and the remaining timing measures. Two further differences are not resolvable at this sample size: number of movement units, 0.84 $[0.75, 0.89]$ against 0.77, and single-trial interjoint coordination, 0.48 $[0.25, 0.82]$ against 0.71, each of their values falling inside our interval. Since [2] reports no intervals, even the two that separate are one-sided: their point estimate lies

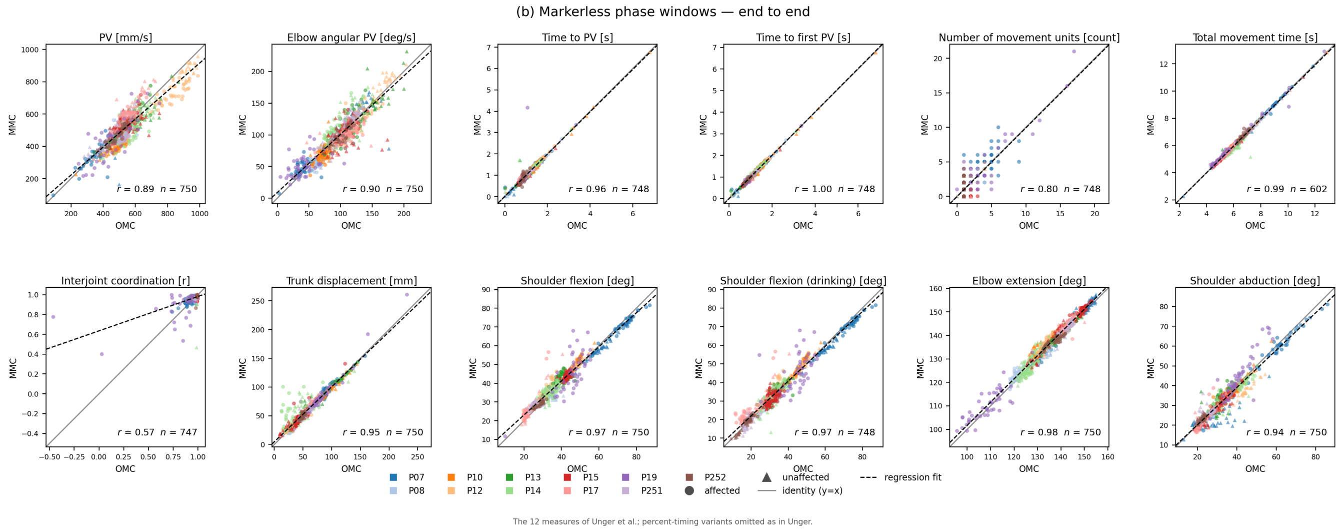

Fig. 2. The twelve movement-quality measures of [2], MMC versus OMC, per participant and arm, with both the pose and the phase windows taken from the markerless recording. Same trials as Table II.

outside our interval, which is weaker than the two estimates being distinguishable from each other.

The comparison is not controlled. The trial subset and the exclusions differ, the movement-unit threshold is 60 mm/s here against the protocol's 20, and our reaching window is 31% shorter than the reference protocol's (Sec. IV-E). That last is the one that bears on these measures directly: all four are reduced over the reaching window, and a shorter window gives spurious speed oscillations less room to be counted, which would flatter the movement-unit count in particular. The threshold does not account for any margin, agreement being flat across that range (Sec. III-D); the window length is not something this comparison can rule out. A geometric construction from keypoints therefore gives up little of the agreement a fitted model achieves. What it buys is not primarily raw speed — both pipelines are dominated by 2D keypoint extraction — but a bounded per-trial cost whose expensive part consumes single frames, so that a front end fast enough to keep up with capture leaves only 1.3 s of the 16.8 once the recording stops.

The two weakest measures are the two that [2] reports as its own weakest, and both are limited by their summary statistic rather than by the reconstruction. Interjoint coordination has almost no variance to track: its optical values span 0.976 to 0.995 across the interquartile range and 94% of their variance sits in 5% of trials, so the correlation reads low even where the systems agree trial by trial — its rank correlation is 0.65 against the Pearson figure of 0.48. Resampling the same trials moves it between 0.27 and 0.98, so differences of the size reported here say nothing about the pipeline. Number of movement units has almost no resolution: three quarters of the optical values are exactly 1, and the two systems agree exactly on 59% of trials and to within one unit on 91%. Both are better reported as agreement within a tolerance. Summarising the flexion–elbow coupling by the shoulder's share of the reach's angular travel instead raises agreement to $r_s = 0.89$ and needs no landmark matching, but what such a reformulation measures clinically is a separate study.

What remains of the disagreement elsewhere is landmark correspondence. Fitting a constant per-landmark offset lifts the maximum joint angles from 0.68–0.90 to 0.94–0.98, so most of what presents as pose error is definitional; the wrist needs 3 mm, so the end-effector was never the difficulty. The offset does not transfer across participants, making it a property of each recording session rather than a fixed difference between the two marker conventions, and which system carries it is not identifiable from this comparison. The fit is also specific to what it was fitted on: tuned to the angle maxima, it degrades a coordination statistic built from the same angles' derivatives. A markerless system in the field has no optical landmarks to fit against, so the measures worth preferring are those that need no such fit, and the fit itself is better made against a biomechanical model than against a second measurement system.

Segmenting from the markerless cup costs almost nothing, and what cost there is follows which window a measure uses rather than how badly a boundary is placed. Only 1.3 s of the 16.8 is work that cannot begin before the last frame, so same-session feedback is a matter of engineering rather than of method: what stands between this pipeline and a live one is a front end that keeps pace with capture, which this card does not quite manage at five cameras and 60 Hz. Halving the capture rate buys that headroom and is close to free, provided the temporal smoother's window is scaled with the rate: the 6–9% loss of peak magnitude that a naive port produces is an artefact of holding a learned filter's window at a fixed sample count, not a cost of sampling. The only cost distinguishable from zero is peak velocity, which loses 0.028; the other eleven measures cannot be separated from no change at all (Sec. IV-F).

VI. Limitations

Because the two systems label different points on the body, their absolute magnitudes differ by amounts specific to each measure: markerless peak velocity runs 3.7% below the optical value and trunk displacement 1.5% above it, and the markerless upper arm is 21% shorter while the forearm agrees within 3%, as the $97\,\mathrm{mm}$ shoulder displacement of Sec. IV-C implies. Absolute values here are distances between two conventions rather than anatomical quantities; agreement is reported as correlation, which the per-session offsets leave intact. Those correlations pool across participants and arms, following [2], and pooling supplies between-participant range that a clinic scoring one patient would not have: shoulder flexion reads 0.97 pooled against 0.85 within a participant and arm (Sec. IV-C).

The cohort is small, from a single site, and its impairment gradient is mild. The pipeline also depends on camera placement and calibration quality: eight of the eleven recording units had to give up one or two of their five cameras before the remainder were usable, and the participant whose calibration is worst over the cameras actually used is the one whose measures agree least well. The two sessions of the participant recorded twice differ sevenfold in calibration error over the same five cameras, which is why they are treated separately throughout. Finally, the joint angles are geometric constructions from keypoints and not anatomical joint angles; they are compared against the same constructions applied to the optical markers, which makes the comparison internally consistent but does not make either quantity a clinical joint angle.

VII. Conclusion

Movement-quality measures for the standardized drinking task can be recovered from multi-camera video without markers, without inverse kinematics and without a biomechanical model, at an agreement with optical motion capture that is within 0.02 of a fitted-model pipeline on most measures, at $16.8\,\mathrm{s}$ per trial against roughly fifteen minutes, and with only $1.3\,\mathrm{s}$ of that unable to begin before the recording ends. The pipeline segments the task itself and reports when it cannot, which is the part the reference comparison leaves open.

Acknowledgment

[TODO: funding, DELTA study data providers]

[TODO: REFERENCES: all ten entries are cited and every field is checked against a publisher record (provenance in refs.bib). ONE gap remains and needs institutional access, not more searching: every number quoted from [2] — their Table III, the 12-day/1160-trial figure, the ∼30 CPU-hour IK figure, and both the Sec. III-D2 and Limitations quotations — was read off arXiv v1, and arXiv has no v2. The published version, IEEE TMRB 8(1):90–97, doi 10.1109/TMRB.2025.3605962, is not open access (Unpaywall, checked 2026-08-25), so it must be fetched through a subscription and those numbers rechecked before submission.]

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